



MIHIR LAKHANI

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PROFESSIONAL SUMMARY

2027 B.Tech Computer Science candidate with hands-on experience building end-to-end machine learning pipelines, explainable AI solutions, real-time digital twins, containerized REST services, and full-stack telemetry platforms. Strong foundation in Python, Docker, Flask, MongoDB, React, Git, and production-oriented model evaluation, with a focused interest in enterprise AI-native applications, RAG pipelines, agentic workflows, and LLM-integrated products.

PROFESSIONAL EXPERIENCE

AI/ML Research Intern - Network Digital Twin Project

Jan 2025 – Jun 2025

- Worked on a faculty-led Network Digital Twin research project exploring SLA-violation risk prediction, explainability, and proactive network-control ideas.
- Integrated SHAP-based explainability and root-cause analysis to make model outputs interpretable and actionable for engineering decisions.
- Built a Random Forest prototype to classify SLA-violation risk from stored network KPI snapshots.
- Won 3rd Prize at Nokia Campus Connect 2025 for the AI-powered network SLA prediction system.

PROJECTS

Source-Cited RAG Assistant: Ask About My Work

Jun 2026 - Aug 2026

- Built a source-cited portfolio assistant that lets visitors explore approved public information about my projects, skills, certifications, and background through natural-language questions.
- Implemented a Flask-based RAG workflow with local hybrid retrieval and Gemini to produce evidence-grounded answers with trusted links and approved diagrams.
- Restricted retrieval to reviewed public sources, validated citations before display, and declined unsupported or private requests rather than guessing.

5G Handover Prediction using Stability-Aware ML

Jan 2026 - Apr 2026

- Developed a Novel proactive mobility decision engine combining network KPIs, signal prediction, trend analysis, and custom Pt/St/Vt/Rt logic to reduce unstable handover triggers.
- Outperformed traditional RSRP-threshold logic by reducing degraded-signal duration by 70 seconds in a 300-second multi-gNB simulation.

Fleet Smart Vehicle Digital Twin

Aug 2025 - Nov 2025

- Engineered a real-time distributed platform to ingest, synchronize, and visualize live fleet telemetry through containerized backend services and a React interface.
- Developed REST APIs, MQTT-based event flows, and modular twin-state services; designed Eclipse Ditto Thing schemas for scalable, cloud-ready deployment.

Parkinson's Disease Detection using ITDO Attention DenseNet

Jul 2025 - Aug 2025

- Implemented an end-to-end EEG pipeline for preprocessing, feature engineering, training, and evaluation; optimized Attention DenseNet with Improved Tasmanian Devil Optimization.
- Achieved an F1-score of 0.89 on unseen neurological signal data.

TECHNICAL SKILLS

AI/ML Engineering: Python, machine learning pipelines, RAG, Agentic AI, model inference, statistical and signal analysis
AI Systems & Backend: REST APIs, Flask, data pipelines, event-driven telemetry, containerized services, Docker, MQTT, MongoDB, MySQL, Eclipse Ditto, n8n
Full-Stack & Engineering: React, HTML, CSS, C/C++, OOPs, Git/GitHub, Jupyter, modular software engineering

EDUCATION

B.Tech, Computer Science - SRM Institute of Science & Technology, Chennai	2023 - 2027
Class XII, Science with Mathematics - St. Paul's Sr. Sec. School (CBSE), Rajasthan	2021 - 2022
Class X, Secondary Education - Maharana Mewar Public School (CBSE), Rajasthan	2019 – 2020

CERTIFICATIONS & ACHIEVEMENTS

Achievements: 3rd Prize - Nokia Campus Connect 2025 | 1st Place - Creative Ingenuity 2023 (AutoCAD Design Competition)
Certifications: OCI 2025 Generative AI Professional (Oracle) | OCI 2025 Certified Data Science Professional (Oracle) | Microsoft AI & ML Engineering (Coursera) | Machine Learning with Python – IBM (Coursera) | Python for Everybody - University of Michigan (Coursera)